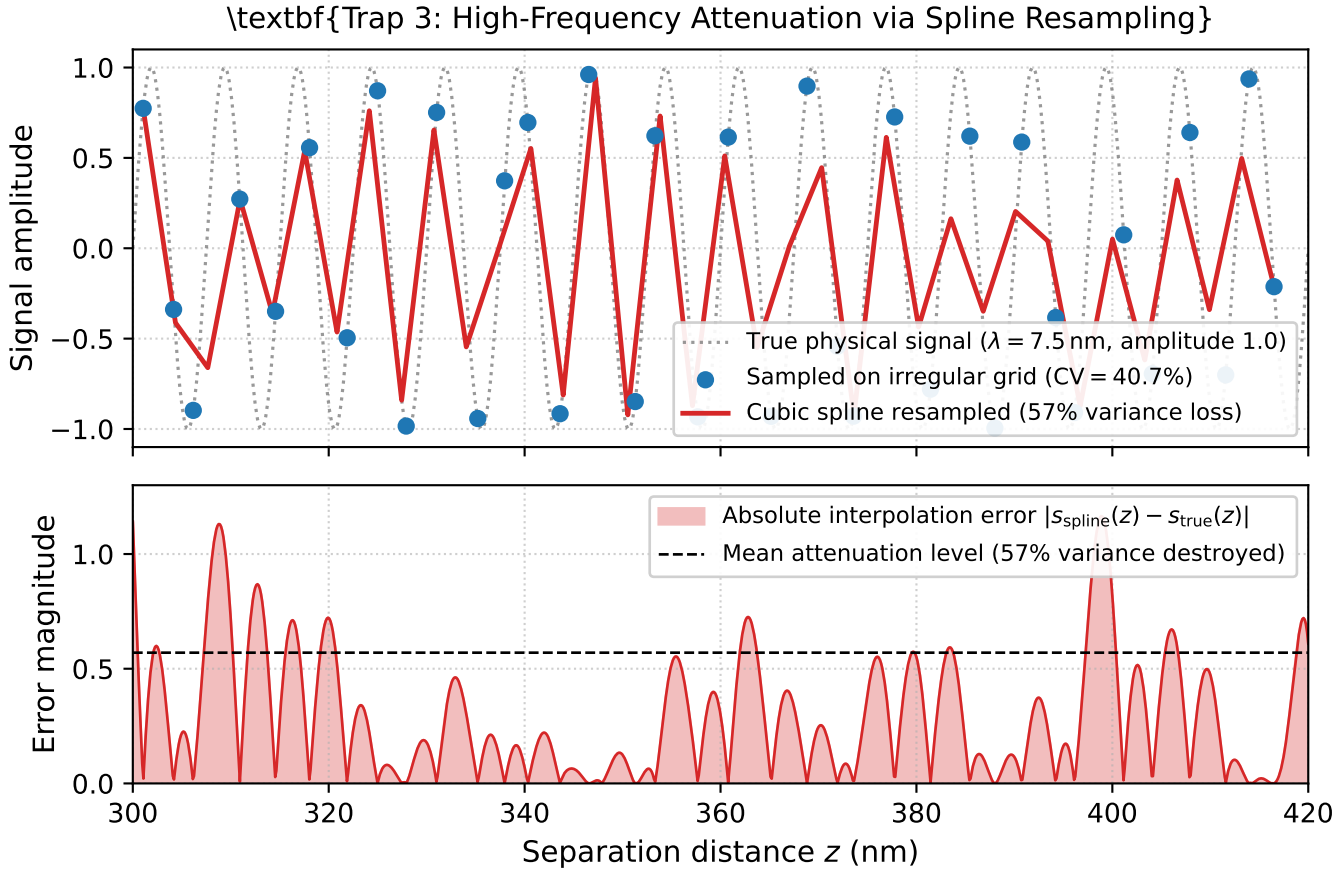
\textbf{Trap 3: High-Frequency Attenuation via Spline Resampling}
Signal amplitude
1.0
0.5
0.0
−0.5
−1.0
True physical signal (λ = 7.5 nm, amplitude 1.0)
Sampled on irregular grid (CV = 40.7%)
Cubic spline resampled (57% variance loss)
Error magnitude
1.0
0.5
0.0
Absolute interpolation error |s_spline(z) − s_true(z)|
Mean attenuation level (57% variance destroyed)
300
320
340
360
380
400
420
Separation distance z (nm)